Heterogeneous Households and Aggregate Dynamics: Analytical Moment Recursion[1]

Online Appendix

Contents

A Proofs **3**

A.1 Proof of Proposition 1 . . . 3

A.2 Proof of Proposition 2 . . . 4

A.3 Proof of Proposition 3 . . . 4

A.4 Proof of Corollary 2 . . . 4

A.5 Proof of Corollaries 3 and 4 . . . 5

A.6 Proof of Theorem 1 . . . 5

A.7 Proof of Proposition 4 . . . 5

B Extended Theoretical Discussion **6**

B.1 Transition matrix interpretation . . . 6

B.2 Signal-to-noise ratios . . . 6

B.3 Special cases: extended discussion . . . 7

B.4 State-dependent inequality dynamics . . . 8

C Asset Pricing Details **9**

C.1 Static and dynamic decomposition . . . 9

C.2 Wealth mobility . . . 10

C.3 Risk premium sensitivity . . . 10

D Epstein-Zin Extension **11**

E Additional Quantitative Results **12**

E.1 Implied TFP comparison . . . 12

E.2 Counterfactual multi-variable panels . . . 12

E.3 Transition dynamics . . . 12

E.4 Distributional state dynamics . . . 13

E.5 Sorting decomposition . . . 13

E.6 Historical distribution of TFP gains . . . 14

[1]Hanbaek Lee, University of Cambridge, Email: hanbaeklee1@gmail.com

E.7 Consumption dynamics 17
E.8 Historical episode analysis 17
E.9 Robustness 18

A Proofs

A.1 Proof of Proposition 1

Proposition 1 (Analytical household-level decisions)**.**
The optimal consumption, saving, and labor supply policies are:

$$c(x;X) = \frac{1-\beta}{1+\theta}(1+r(X)(1-\tau)e)a + \left(\frac{1-\beta}{1+\theta}a\xi + \kappa\right) w(X)(1-\tau)z \tag{1}$$

$$a'(x;X) = \beta a\left((1+r(X)(1-\tau)e) + w(X)(1-\tau)z\xi\right) + T(X) \tag{2}$$

$$n(x;X) = \max\left\{\kappa(1+\theta) + \xi a - \frac{\theta\, c(x;X)}{w(X)(1-\tau)z},\ 0\right\} \tag{3}$$

Proof.
I assume the constraint $a' \geq 0$ does not bind and later verify that the solution satisfies this. From the first-order conditions,

$$[c]: \quad \frac{1}{c - \kappa w(X)(1-\tau)z} = \lambda \tag{4}$$

$$[a']: \quad v_a(x';X') = \lambda \tag{5}$$

$$[n]: \quad \frac{\theta}{\kappa + \xi a - n} = \lambda w(X)(1-\tau)z. \tag{6}$$

From the envelope condition,

$$[a]: \quad v_a(x;X) = \frac{\theta\xi}{\kappa + \xi a - n} + \lambda(1 + r(X)(1-\tau)e). \tag{7}$$

From Equations (4) and (6),

$$\frac{\theta}{\kappa + \xi a - n} = \frac{w(X)(1-\tau)z}{c - \kappa w(X)(1-\tau)z}. \tag{8}$$

Rearranging,

$$n(x;X) = \kappa(1+\theta) + \xi a - \theta\frac{c(x;X)}{w(X)(1-\tau)z}. \tag{9}$$

The budget constraint can be written as:

$$(1+\theta)(c - \kappa w(X)(1-\tau)z) + a' = ((1 + r(X)(1-\tau)e) + w(X)(1-\tau)z\xi)a + T(X). \tag{10}$$

The transfer $T(X)$ is identical across households and enters savings as an additive intercept. The result of Brock and Mirman (1972) immediately applies to the multiplicative component. ■

A.2 Proof of Proposition 2

Proof sketch. The saving rate $s(x; X) = (a'(x; X) - a)/y(x; X)$ inherits the properties of the saving policy a'. Property (ii) follows immediately from $s(0; X) = T/(\kappa\theta w(X)(1-\tau)z+T) > 0$. Property (i) follows from taking the limit $a \to \infty$, under which the transfer T becomes negligible. Property (iii)—monotonicity and concavity—follows from direct differentiation: $\partial s/\partial a > 0$ because the multiplicative component of a' grows faster than income, and $\partial^2 s/\partial a^2 < 0$ because the subsistence floor absorbs a diminishing share of resources. ■

A.3 Proof of Proposition 3

Proof. The saving policy $a'(x; X) = \beta a((1 + \tilde{r}e) + \tilde{w}\xi z) + T$ is linear in a with coefficients that depend on (z, e) and current prices. Aggregating across the distribution:

For aggregate capital: $K' = \int a'(x; X)\, d\Phi = \beta[(1+\tilde{r}+\tilde{w}\xi)K+\tilde{r}\,\mathrm{cov}(e, a)+\tilde{w}\xi\,\mathrm{cov}(z, a)]+T$, where we used $\mathbb{E}[e] = \mathbb{E}[z] = 1$.

For $\mathrm{cov}(z', a')$: since $z' = z^{\rho_z}\exp(\varepsilon'_z)$ and a' is linear in a, we compute $\mathbb{E}[z'a'] - \mathbb{E}[z']\mathbb{E}[a']$. The innovation ε'_z is independent of current states, so $\mathrm{cov}(z', a') = \rho_z\phi_z\beta[\tilde{w}\xi\sigma_z^2 K + (1 + \tilde{r} + \tilde{w}\xi)\mathrm{cov}(z, a)]$, where $\phi_z = \mathrm{var}(z)/(\mathrm{var}(z) + \sigma^2_{\varepsilon,z})$ is the signal-to-noise ratio.

An analogous computation yields the recursion for $\mathrm{cov}(e', a')$, and the independence of z and e produces the zero off-diagonal entries. The system closes because prices (r, w) depend on Φ only through $(K, \mathrm{cov}(z, a), \mathrm{cov}(e, a))$ via the labor supply equation, which itself depends only on these three moments. Uniqueness follows from the contraction property of the moment map when $\rho(\beta\tilde{\mathcal{G}}) < 1$. ■

A.4 Proof of Corollary 2

Proof. The system $\mathcal{M}' = \beta\tilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is a non-homogeneous linear recursion. When the spectral radius $\rho(\beta\tilde{\mathcal{G}}) < 1$, the matrix $I - \beta\tilde{\mathcal{G}}$ is invertible and the unique fixed point $\mathcal{M}^* = (I - \beta\tilde{\mathcal{G}})^{-1}[T, 0, 0]'$ is globally attracting. This follows from the standard theory of linear difference equations: the general solution is $\mathcal{M}_t = (\beta\tilde{\mathcal{G}})^t(\mathcal{M}_0 - \mathcal{M}^*) + \mathcal{M}^*$, which converges to $\mathcal{M}^*$ as $t \to \infty$ whenever all eigenvalues of $\beta\tilde{\mathcal{G}}$ lie inside the unit circle. The additive component $T > 0$ is essential: it places the system in the class of Kesten processes (Kesten, 1973), which require both multiplicative and additive components to generate a

well-defined stationary distribution with Pareto tails. Without T, the system is purely multiplicative and does not admit a non-degenerate stationary solution. ■

A.5 Proof of Corollaries 3 and 4

Proof of Corollary 3. Set $e = 1$ throughout. Then $\mathrm{cov}(e, a) = 0$ for all t, the third row and column of $\tilde{\mathcal{G}}$ become irrelevant, and the system reduces to the 2×2 recursion stated. ■

Proof of Corollary 4. Set $e = 1$ and $z = 1$. Then both covariances vanish identically, and the system reduces to a single equation $K' = \beta(1 + \tilde{r} + \tilde{w}\xi)K + T$. The aggregate consumption function follows from $C = Y + (1 - \delta)K - K'$. ■

A.6 Proof of Theorem 1

Proof. The saving policy $a'(x; X) = \beta a((1+\tilde{r}e)+\tilde{w}\xi z)+T$ depends only on current after-tax prices $(\tilde{r}, \tilde{w})$, the current transfer T, and the household's current state (a, z, e). None of these objects involves expectations of future aggregate states. The consumption policy $c(x; X)$ and labor supply $n(x; X)$ inherit this property. Since all individual policies are independent of beliefs about future aggregates, the aggregate law of motion $\mathcal{M}' = \beta\tilde{\mathcal{G}}\mathcal{M} + [T, 0, 0]'$ is also independent of beliefs: it depends only on current prices (which are functions of the current state $\mathcal{M}$) and the current transfer.

The economic mechanism is the certainty-equivalence property of log preferences: under $u = \log(c - g)$, the income and substitution effects of changes in future returns exactly cancel, so the household's saving decision depends only on current resources and prices. This household-level property aggregates up because the linear policy functions preserve the moment structure. ■

A.7 Proof of Proposition 4

Proof. The BRCE requires that household policies be optimal given *arbitrary* beliefs about aggregate dynamics. Theorem 1 establishes that the saving policy is belief-invariant. Since the aggregate law of motion depends only on the collection of saving policies (through the moment aggregation), and since these policies are identical regardless of beliefs, the BRCE law of motion coincides with the MRCE law of motion. Uniqueness of the MRCE then implies uniqueness of the BRCE, and the allocations are identical. ■

B Extended Theoretical Discussion

B.1 Transition matrix interpretation

The state vector of the MRCE has three components. The first, K, is aggregate capital—the mean of the wealth distribution. This is the single state variable that Krusell and Smith (1998) found to be nearly sufficient in their numerical experiments. The second, $\text{cov}(z, a)$, measures the alignment between labor productivity and wealth: it is positive when productive workers are also wealthy, and zero when productivity and wealth are uncorrelated. The third, $\text{cov}(e, a)$, measures the alignment between return heterogeneity and wealth: it is positive when high-return households are also wealthy.

The law of motion takes the compact form $\mathcal{M}' = \beta\,\widetilde{\mathcal{G}}\,\mathcal{M} + [T, 0, 0]'$, where $\widetilde{\mathcal{G}}$ is a 3×3 matrix that depends on current after-tax prices. This is a non-homogeneous multivariate linear recursion, with a multiplicative component discounted by β and an additive intercept from the transfer T. The diagonal entries of $\widetilde{\mathcal{G}}$ show that each moment feeds into its own future value—capital tomorrow depends on capital today, and each covariance depends on its own lagged value. The off-diagonal entries capture the cross-moment interactions: for instance, the $(1, 3)$ entry $\tilde{r}$ shows that the covariance between returns and wealth contributes to future capital through the extra saving of high-return, wealthy households. The $(3, 1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that higher aggregate capital generates a higher return, which in turn strengthens the sorting between returns and wealth.

The zero entries in the matrix are also informative. The $(2, 3)$ and $(3, 2)$ entries are both zero, meaning that the two covariances do not directly interact with each other. Each covariance interacts with aggregate capital, but the labor-productivity-wealth alignment and the return-wealth alignment evolve as separate channels. This clean separation has a deep economic reason: labor productivity z and return heterogeneity e are assumed to be independent processes. Since z and e do not directly influence each other at the household level, their respective covariances with wealth cannot influence each other at the aggregate level either. The two sorting mechanisms—productive workers becoming wealthier, and high-return households accumulating more—operate through the common channel of aggregate capital but do not contaminate each other.

B.2 Signal-to-noise ratios

The parameters ϕ_z and ϕ_e—the signal-to-noise ratios of the idiosyncratic processes—deserve special attention. These parameters control the rate at which idiosyncratic shocks "reshuffle" households across the distribution. A high ϕ (close to one) means that shocks are highly

persistent and reshuffling is slow: households that are productive today are very likely to be productive tomorrow, and the sorting between productivity and wealth strengthens over time. A low ϕ (close to zero) means rapid reshuffling: today's high earners may be tomorrow's low earners, and the covariance between productivity and wealth dissipates quickly. In the transition matrix, ϕ appears as a scaling factor on the off-diagonal entries, controlling the speed at which the distributional state evolves. This creates a natural distinction between "fast" and "slow" distributional dynamics: economies with persistent idiosyncratic shocks exhibit slow-moving distributional states that interact strongly with aggregate fluctuations, while economies with transitory shocks have distributional states that adjust quickly and play a smaller role in aggregate dynamics.

B.3 Special cases: extended discussion

Approximate aggregation. When there is no return heterogeneity ($e = 1$), the economy is governed by just two state variables: aggregate capital and the covariance between productivity and wealth. This is the analytical counterpart of the Krusell–Smith economy, enriched by the additional covariance state. The covariance $\mathrm{cov}(z, a)$ captures the "sorting" between productivity and wealth that is present in the data but absent in the Krusell–Smith model when approximate aggregation holds.

This special case illuminates the relationship between the present paper and the approximate aggregation literature. Krusell and Smith (1998) found numerically that the mean of the wealth distribution is "nearly sufficient" to forecast future prices in a calibrated economy with only labor income risk. The present result shows that, in a model with similar features, the mean is *exactly* sufficient if and only if the covariance $\mathrm{cov}(z, a) = 0$—that is, if productivity and wealth are uncorrelated. In practice, the covariance is small relative to K in many calibrations, which explains why approximate aggregation works well numerically: the error from ignoring $\mathrm{cov}(z, a)$ is proportional to the magnitude of this covariance, which is indeed small when productivity shocks are not too persistent or too dispersed. But the error is not zero, and it grows with the persistence and dispersion of idiosyncratic productivity—precisely the conditions under which the Krusell–Smith approximation is known to break down.

Representative-agent nesting. When all household heterogeneity is eliminated, both covariances vanish, and the model reduces to a version of the stochastic growth model with log preferences and endogenous labor supply. The aggregate consumption function in the representative-agent case, $C = \frac{1-\beta}{1+\theta}(1 + \tilde{r} + \tilde{w}\xi)K + \kappa\tilde{w}$, has two components: a "permanent income" term proportional to aggregate wealth and a "subsistence" term proportional to

the after-tax wage. The heterogeneous-agent model adds distributional corrections to both terms, and the quantitative section shows that these corrections, while small in normal times, can become economically significant during episodes of high inequality or large distributional shifts.

B.4 State-dependent inequality dynamics

The moment recursion pins down how inequality evolves over time. The transition matrix $\widetilde{\mathcal{G}}$ makes these feedback effects precise. The $(3,1)$ entry $\phi_e \rho_e \tilde{r} \sigma_e^2$ shows that a higher aggregate capital stock—which raises the after-tax return—strengthens the covariance between returns and wealth. The $(2,1)$ entry $\phi_z \rho_z \tilde{w} \xi \sigma_z^2$ shows a parallel effect for the productivity-wealth covariance. These off-diagonal interactions generate endogenous amplification: a positive productivity shock raises capital, which raises returns, which strengthens the return-wealth sorting, which in turn affects future capital accumulation.

This feedback loop has a natural economic interpretation in terms of "wealth spirals." When a positive aggregate shock raises returns, households that already hold large portfolios benefit the most. Their elevated income is partially saved, increasing their wealth further. Because the return process is persistent (ρ_e close to one), these same households are likely to enjoy high returns again next period, reinforcing the sorting between wealth and returns. Over time, this mechanism generates increasing concentration of wealth among high-return households—a dynamic that is consistent with the secular rise in top wealth shares documented by Piketty (2014) and Saez and Zucman (2016). The MRCE makes this mechanism analytically precise: the rate at which wealth concentrates is governed by the dominant eigenvalue of $\beta\widetilde{\mathcal{G}}$, and the steady-state level of concentration is pinned down by the fixed point $\mathcal{M}^*$.

Conversely, the model also identifies forces that limit concentration. The discount factor $\beta < 1$ ensures that a fraction $1-\beta$ of each household's wealth is consumed each period, providing a force toward deconcentration. The transfer T further compresses the distribution by providing an additive floor on savings that benefits low-wealth households disproportionately. The steady-state wealth distribution is the outcome of these competing forces: the multiplicative amplification from return sorting pushes toward concentration, while discounting and transfers push toward equality.

C Asset Pricing Details

The model delivers a novel result for asset pricing: individual return heterogeneity, which is traditionally viewed as a diversifiable idiosyncratic risk, carries a positive risk premium because it has aggregate consequences. This result connects the model to a longstanding debate in finance about whether idiosyncratic risk is priced (Constantinides and Duffie, 1996; Storesletten et al., 2004).

In the standard complete-markets framework, idiosyncratic risk washes out in the aggregate by the law of large numbers. But in the present model, idiosyncratic return heterogeneity does *not* wash out, because it creates systematic wealth sorting that alters the composition of the economy. Households differ in the returns they earn on their savings, captured by the idiosyncratic factor e. Over time, households with persistently high e accumulate more wealth than households with low e—this is the return-sorting channel formalized by the covariance $\text{cov}(e, a)$. This wealth sorting changes the composition of the economy: wealthy households have different labor supply behavior than poor households (they work less due to the wealth effect). As the composition shifts, aggregate labor supply changes, which changes aggregate returns for *everyone*. Because individual return heterogeneity e affects aggregate outcomes through the wealth-sorting channel, it is not fully diversifiable.

C.1 Static and dynamic decomposition

The asset pricing implications decompose into a static and a dynamic component.

Static effect. Holding the aggregate state fixed, an increase in the return-wealth covariance $\text{cov}(e, a)$ reduces the current aggregate return:

$$\frac{\partial r(\mathcal{M}, A)}{\partial \text{cov}(e, a)} = \frac{\partial r}{\partial L} \frac{\partial L}{\partial \text{cov}(e, a)} < 0 \tag{11}$$

The logic is that a higher covariance strengthens the wealth effect on labor supply, reducing employment and output, and thereby reducing the marginal product of capital.

Dynamic effect. Looking one period ahead, the effect is reinforced:

$$\frac{\partial r(\mathcal{M}', A')}{\partial \text{cov}(e, a)} = \frac{\partial r'}{\partial K'} \frac{\partial K'}{\partial \text{cov}(e, a)} + \frac{\partial r'}{\partial \text{cov}(e', a')} \frac{\partial \text{cov}(e', a')}{\partial \text{cov}(e, a)} < 0 \tag{12}$$

Both terms are negative. The first says that a higher return-wealth covariance today raises capital accumulation (because high-return, wealthy households save more), which depresses

future returns through diminishing marginal products. The second says that the return-wealth covariance is persistent—today's sorting propagates into tomorrow's sorting—and continues to depress returns through the labor supply channel.

C.2 Wealth mobility

The model provides an analytical expression for the half-life of wealth—the time it takes for a household's wealth to converge halfway to the cross-sectional mean:

$$T(\mathcal{M}, A) = \frac{\log(2)}{\log(\beta(1 + r(\mathcal{M}, A)(1 - \tau)))} \tag{13}$$

This expression highlights a tension between two forces. Faster mean reversion of the return-wealth covariance implies quicker adjustment of the distributional state, but slow rank mobility in the wealth distribution means that the wealth positions of individual households change only gradually. The combination generates persistent effects of idiosyncratic return shocks on aggregate outcomes, even when the cross-sectional covariance itself adjusts relatively quickly.

Taken together, the asset pricing results suggest that models with heterogeneous returns to wealth—such as those motivated by the empirical findings of Fagereng et al. (2020) and Bach et al. (2020)—have implications not just for the wealth distribution but also for asset prices and aggregate dynamics.

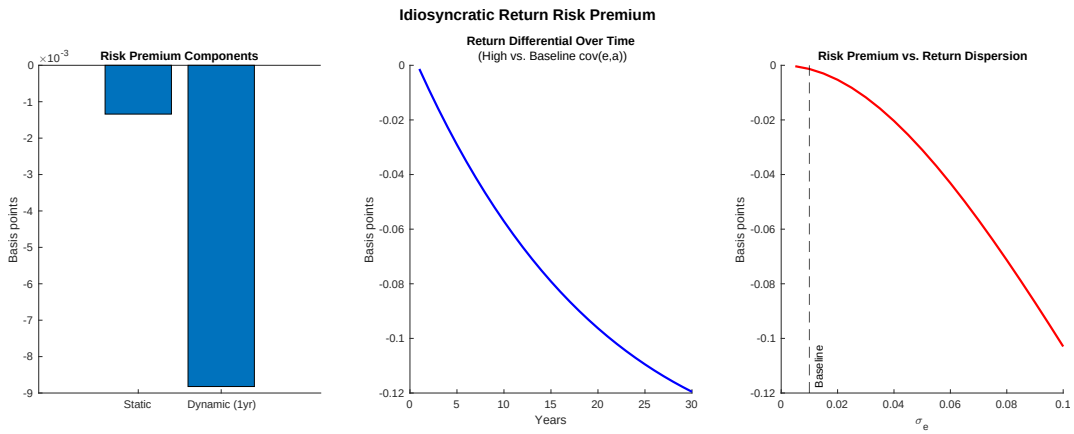
C.3 Risk premium sensitivity

The main text establishes that higher $\text{cov}(e, a)$ lowers the equilibrium interest rate. To quantify this channel, we compare steady-state economies at different levels of return dispersion σ_e. At the baseline calibration ($\sigma_e = 0.01$), the interest rate effect is -0.001 basis points—negligible. The premium grows quadratically with σ_e:

$$\underbrace{-0.03 \text{ bp}}_{\sigma_e=0.05} \quad \underbrace{-0.10 \text{ bp}}_{\sigma_e=0.10} \quad \underbrace{-0.70 \text{ bp}}_{\sigma_e=0.50} \quad \underbrace{-1.01 \text{ bp}}_{\sigma_e=0.83}$$

The threshold for a 1 basis point premium requires $\sigma_e \approx 0.83$. At empirically realistic σ_e, the main consequence of return heterogeneity operates through the *amplification channel* (state-dependent impulse responses), not the *pricing channel*.

Figure C.1: Return risk premium sensitivity. Left: static vs. dynamic premium components at steady state. Center: cumulative return differential over 30 years. Right: premium as a function of return dispersion σ_e.



D Epstein-Zin Extension

Under Epstein-Zin preferences with unit elasticity of intertemporal substitution (EIS = 1), the flow composite is:

$$\tilde{c} \equiv c - \kappa z w(X)(1-\tau), \qquad m \equiv (\kappa + \xi a) - n, \qquad X^{\text{comp}} \equiv \tilde{c}\, m^{\theta}.$$

The intratemporal optimality condition implies $m = (\theta/(w(X)(1-\tau)z))\tilde{c}$, so:

$$X^{\text{comp}} = \tilde{c}^{\,1+\theta} \left(\frac{\theta}{w(X)(1-\tau)z} \right)^{\theta}.$$

The recursive value function satisfies:

$$\log V_t = (1-\beta)\log X_t + \beta \log \left(\mathbb{E}_t \left[V_{t+1}^{1-\gamma} \right] \right)^{\frac{1}{1-\gamma}}, \qquad \beta \in (0,1), \quad \gamma > 0.$$

The critical result is that, with EIS = 1, the optimal saving share $s_a(s)$ satisfies the same type of fixed-point condition as in the log case:

$$s_a(s) = \left[1 + \left(\frac{\beta(1+\theta)^p\, \bar{B}(s)}{C(s)} \right)^{\frac{1}{p-1}} \right]^{-1},$$

where $p \equiv (1+\theta)(1-\sigma)$, $C(s) \equiv (\theta/(w(X)(1-\tau)z))^{\theta(1-\sigma)}$, and $\bar{B}(s) \equiv \mathbb{E}[B(s') \mid s]$ is a

conditional expectation computed from the fixed-point system:

$$B(s) = \Pi(s)^p \left[C(s) \left(\frac{1 - s_a(s)}{1 + \theta} \right)^p + \beta \, s_a(s)^p \, \bar{B}(s) \right],$$

with $\Pi(s) \equiv (1 + r(X)(1 - \tau))e + w(X)(1 - \tau)z\xi$.

The MRCE and BRCE continue to hold under Epstein-Zin preferences with unit EIS. The saving share is still independent of the household's wealth level, and the law of motion retains its closed-form matrix structure. The parameter γ controls the household's aversion to risk over consumption *lotteries* (a cross-sectional concept), while β and θ control the intertemporal and intratemporal tradeoffs (time-series concepts). Separating these roles allows the model to match both the equity premium (through γ) and the saving rate (through β) without forcing them to be determined by a single parameter.

This extension is particularly valuable for asset pricing: with $\gamma \gg 1$, the same wealth-sorting mechanism generates a substantially larger risk premium than under log utility ($\gamma = 1$).

E Additional Quantitative Results

E.1 Implied TFP comparison

Figure E.1 compares the implied TFP series from the MRCE (which accounts for distributional covariances) with the TFP implied by a representative-agent version of the model (which sets all covariances to zero). The two series track each other closely at business cycle frequencies, but the HA model implies a slightly different TFP level, reflecting the contribution of distributional sorting to aggregate labor supply. The figure also shows the Fernald utilization-adjusted TFP series as an external benchmark.

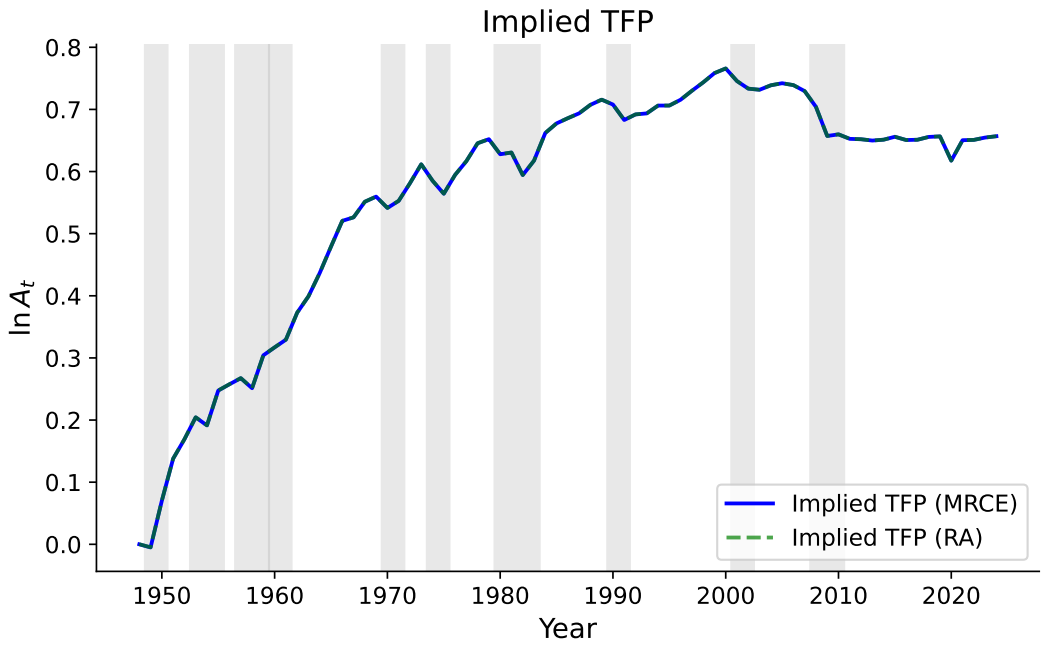
E.2 Counterfactual multi-variable panels

Figure E.2 extends the counterfactual comparison to four macroeconomic variables: output, labor supply, the interest rate, and the wage. TFP is the dominant driver of wages, while the discount factor is the primary driver of cyclical variation in labor supply.

E.3 Transition dynamics

Figure E.3 plots the dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$ over time. The dominant eigenvalue has mean 1.027 and standard deviation 0.013, and exceeds unity in 97.4 percent of sample years.

Figure E.1: Implied TFP from the MRCE compared to the representative-agent analogue and the Fernald utilization-adjusted TFP series. Shaded areas indicate NBER recession dates.



During the rare stable episodes (eigenvalue < 1), the implied half-life of the distributional state averages 64.8 years, underscoring the extreme persistence of distributional dynamics. The near-unit-root behavior generates endogenous persistence in aggregate dynamics that converts transitory shocks into persistent output movements.

E.4 Distributional state dynamics

Figure E.4 plots the two covariances and their capital-normalized counterparts. The productivity-wealth covariance $\text{cov}_t(z, a)$ grows from 0.005 in 1948 to 0.343 in 2024, with a mean annual growth rate of 5.5 percent. The return-wealth covariance grows from 0.001 to 0.009. Both are strongly procyclical. The normalized ratio $\text{cov}(e, a)/K$ remains nearly constant, reflecting the near-unit-root dynamics of e.

E.5 Sorting decomposition

Figure E.5 reports the analytical decomposition of distributional sorting into innovation and persistence channels. Persistence dominates: on average, 85 percent of $\text{cov}(z, a)$ growth comes from the persistence channel. The right panel confirms that the new sorting rate $\widetilde{\mathcal{G}}_{21}$ is strongly positively correlated with TFP, providing a structural foundation for the

Figure E.2: Counterfactual comparison across output (log), labor supply, interest rate, and wage. Each panel compares the baseline with three counterfactual scenarios. Shaded areas indicate NBER recession dates.

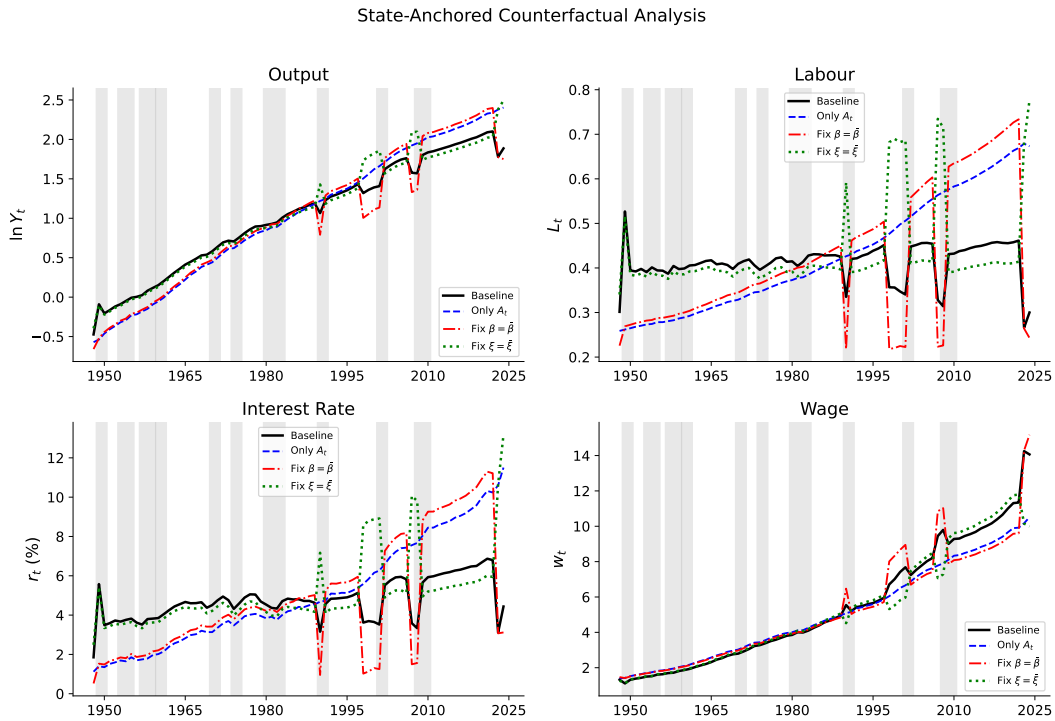
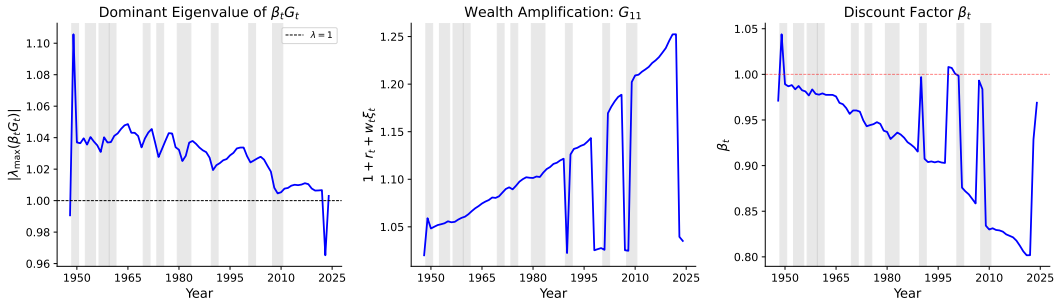


Figure E.3: Transition dynamics. Top: dominant eigenvalue of $\beta_t \widetilde{\mathcal{G}}_t$. Middle: wealth amplification factor. Bottom: time-varying discount factor. Shaded areas indicate NBER recession dates.



TFP-inequality nexus.

E.6 Historical distribution of TFP gains

Table E.1 reports decade averages of TFP gain shares by wealth decile. Figure E.6 visualizes the evolving distribution. The top decile captures approximately 70 percent of TFP-driven wealth gains throughout the sample.

Figure E.4: Distributional state dynamics, 1948–2024. Top row: raw covariances. Bottom row: capital-normalized ratios. Shaded areas indicate NBER recession dates.

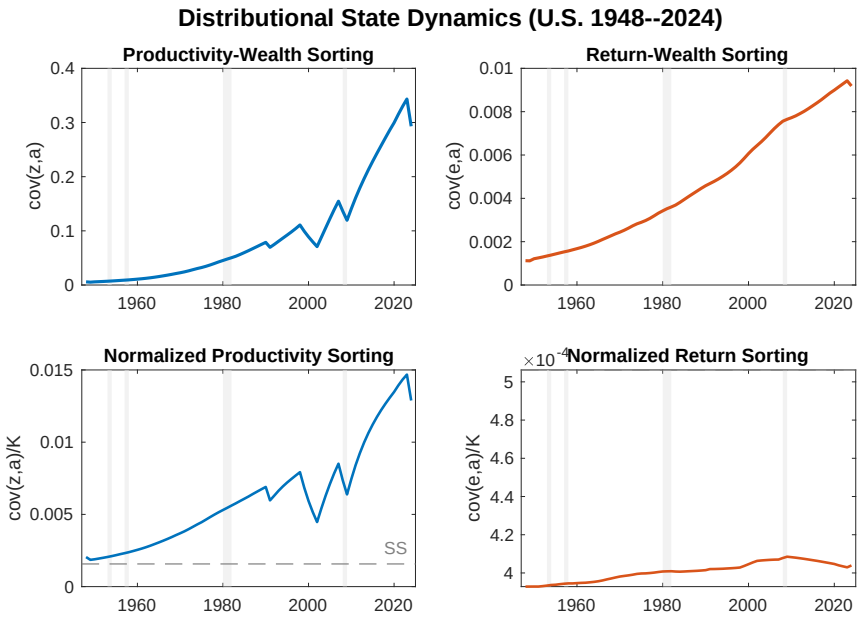


Figure E.5: Analytical decomposition of distributional sorting. Left: innovation and persistence contributions. Center: sorting rate time paths. Right: TFP vs. sorting rate scatter.

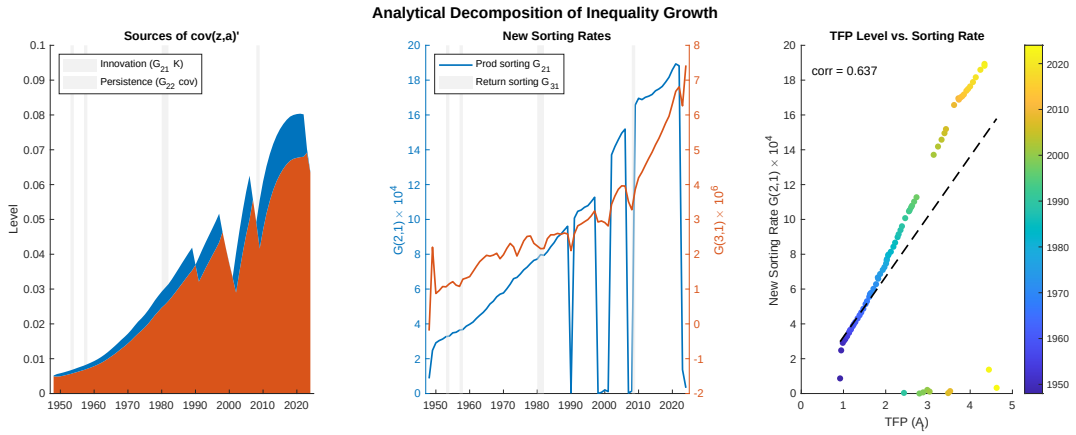
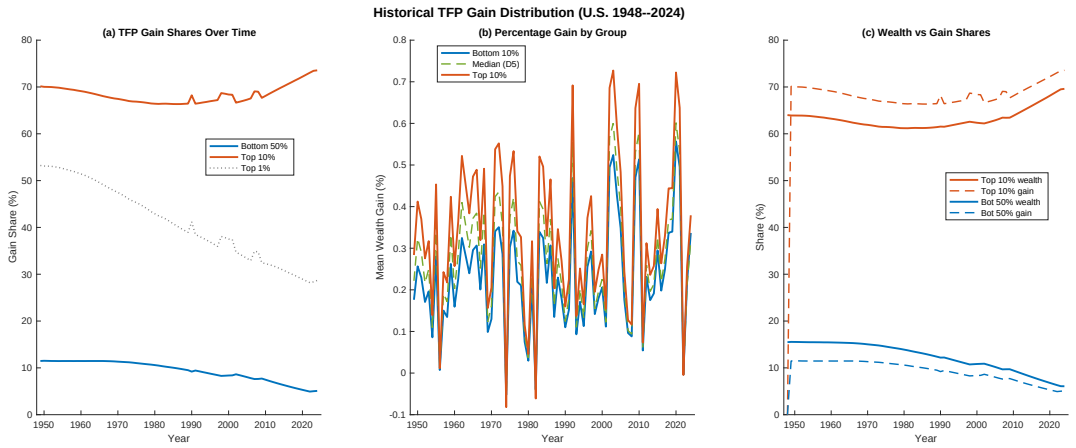


Table E.1: Share of TFP-driven wealth gain by wealth decile (%), decade averages

Decade	D1	D2	D3	D4	D5	D6	D7	D8	D9	D10	Top 5%	Top 1%
1950s	1.6	2.0	2.3	2.6	3.0	3.4	4.0	4.8	6.6	69.7	64.4	52.6
1960s	1.5	1.9	2.3	2.7	3.1	3.5	4.2	5.2	7.2	68.3	62.5	49.7
1970s	1.3	1.8	2.2	2.6	3.1	3.7	4.5	5.7	8.2	66.9	60.2	45.5
1980s	1.1	1.6	2.0	2.4	3.0	3.6	4.6	6.1	9.2	66.4	58.4	41.0
1990s	0.9	1.3	1.7	2.2	2.7	3.4	4.4	6.1	9.8	67.3	58.3	37.8
2000s	0.8	1.2	1.6	2.0	2.5	3.3	4.4	6.2	10.3	67.8	58.0	34.6
2010s	0.6	0.9	1.2	1.6	2.1	2.9	4.0	6.0	10.7	70.0	58.9	30.9
2020s	0.4	0.7	1.0	1.3	1.7	2.4	3.5	5.5	10.6	73.0	61.2	28.6
Full sample	1.1	1.5	1.8	2.2	2.7	3.3	4.2	5.7	9.0	68.4	60.2	41.0

Notes: Each entry reports the average share (in %) of aggregate TFP-driven wealth gain captured by the indicated wealth decile. Period-by-period counterfactual at each year t.

Figure E.6: Historical distribution of TFP gains across wealth percentiles, U.S. 1948–2024.



E.7 Consumption dynamics

The MRCE delivers closed-form aggregate consumption:

$$C_t = \frac{1-\beta_t}{1+\theta}\left[(1+\tilde{r}_t+\tilde{w}_t\xi_t)K_t + \tilde{r}_t\,\mathrm{cov}_t(e,a) + \tilde{w}_t\xi_t\,\mathrm{cov}_t(z,a)\right] + \kappa\tilde{w}_t.$$

Constant marginal propensity to consume. The individual household's MPC out of a one-time transfer is $(1-\beta_t)/(1+\theta)$, which depends on the discount factor and the leisure weight but not on the household's wealth level. At the steady-state calibration ($\beta = 0.96$, $\theta = 2$), the annual MPC is 1.3 percent. The cumulative MPC reaches 6.6 percent after five years and 26.3 percent after twenty years. The constant MPC is a direct consequence of the linear policy functions. This stands in contrast to models with concave consumption functions, where borrowing-constrained "hand-to-mouth" households exhibit MPCs of 0.3–0.5 (Kaplan et al., 2018).

Procyclical MPC. The time-varying MPC $(1-\beta_t)/(1+\theta)$ is procyclical, averaging 2.7 percent during expansions versus 1.2 percent during recessions, providing an additional stabilizing mechanism.

E.8 Historical episode analysis

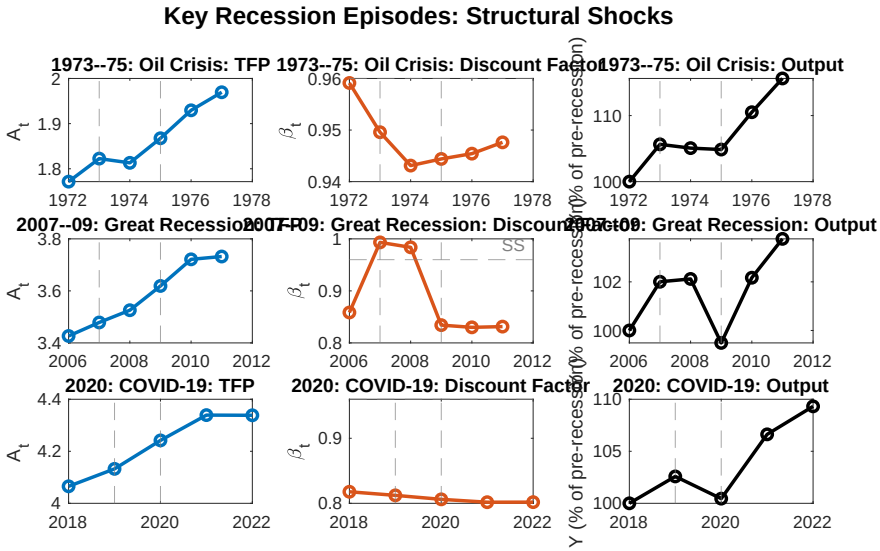
The exact inversion provides a structural narrative for each recession. Figure E.7 displays TFP and the discount factor around three key recessions.

1973–75 (Oil Crisis). Output declined by 0.75 percent from peak to trough while TFP continued to grow (at 1.2 percent per year). The discount factor was essentially unchanged ($\Delta\beta = -0.005$). This is consistent with a supply-side recession.

2007–09 (Great Recession). Output declined by 2.5 percent. TFP continued to grow at trend (2.0 percent per year). The discount factor peaked at 0.993 in 2007 and dropped to 0.834 by 2009—a swing of 0.159. This is consistent with a demand-driven recession.

2020 (COVID-19). Output declined by 2.1 percent. TFP grew above trend (2.6 percent). The discount factor changed minimally ($\Delta\beta = -0.006$). The pandemic was a sui generis shock.

Figure E.7: Structural shocks around three key recessions. Each row shows one episode. Left: TFP. Center: discount factor. Right: indexed output.



E.9 Robustness

Table E.2 reports the sensitivity of the model's key steady-state properties to alternative calibrations. The approximate aggregation result is extremely robust: I^{rep} accounts for 93–99 percent of total investment across all configurations. The near-unit-root persistence is

robust to all parameters except the transfer share η.

Table E.2: Sensitivity of steady-state properties to alternative calibrations

Parameter	Value	K/Y	r (%)	$\lambda_{\max}$	I^{rep} (%)	Half-life (yr)
Baseline		3.23	3.22	0.9988	98.2	563
alpha	0.3	2.89	3.39	0.9986	98.0	503
	0.36	3.61	2.98	0.9989	98.4	629
	0.4	4.23	2.46	0.9991	98.6	736
sigma$_e$	0.001	3.23	3.22	0.9988	98.2	561
	0.005	3.23	3.22	0.9988	98.2	561
	0.02	3.23	3.22	0.9988	98.1	569
	0.05	3.24	3.18	0.9989	97.8	605
	0.1	3.27	3.08	0.9990	97.0	686
theta	1	3.23	3.22	0.9988	98.2	563
	1.5	3.23	3.22	0.9988	98.2	563
	3	3.23	3.22	0.9988	98.2	563
	5	3.23	3.22	0.9988	98.2	563
eta	0.005	3.19	3.35	0.9995	99.3	1480
	0.01	3.21	3.27	0.9991	98.6	745
	0.02	3.26	3.12	0.9982	97.3	378
	0.03	3.31	2.97	0.9973	96.1	255
	0.05	3.41	2.68	0.9956	93.7	157
beta	0.94	2.42	6.65	0.9984	97.6	425
	0.95	2.76	4.94	0.9986	97.9	484
	0.97	3.89	1.47	0.9990	98.5	677
	0.98	4.96	-0.35	0.9992	98.8	862
rho$_z$	0.9	3.23	3.22	0.9988	98.2	562
	0.93	3.23	3.22	0.9988	98.2	563
	0.97	3.23	3.22	0.9988	98.2	563
	0.99	3.23	3.20	0.9988	98.0	566

Notes: Each row varies one parameter from the baseline calibration.

References

Bach, L., L. E. Calvet, and P. Sodini (2020). Rich pickings? Risk, return, and skill in household wealth. *American Economic Review 110*(9), 2703–2747.

Brock, W. A. and L. J. Mirman (1972). Optimal economic growth and uncertainty: The discounted case. *Journal of Economic Theory 4*(3), 479–513.

Constantinides, G. M. and D. Duffie (1996). Asset pricing with heterogeneous consumers. *Journal of Political Economy 104*(2), 219–240.

Fagereng, A., L. Guiso, D. Malacrino, and L. Pistaferri (2020). Heterogeneity and persistence in returns to wealth. *Econometrica 88*(1), 115–170.

Kaplan, G., B. Moll, and G. L. Violante (2018). Monetary policy according to HANK. *American Economic Review 108*(3), 697–743.

Kesten, H. (1973). Random difference equations and renewal theory for products of random matrices. *Acta Mathematica 131*, 207–248.

Piketty, T. (2014). *Capital in the Twenty-First Century*. Cambridge, MA: Harvard University Press.

Saez, E. and G. Zucman (2016). Wealth inequality in the United States since 1913: Evidence from capitalized income tax data. *Quarterly Journal of Economics 131*(2), 519–578.

Storesletten, K., C. I. Telmer, and A. Yaron (2004). Cyclical dynamics in idiosyncratic labor market risk. *Journal of Political Economy 112*(3), 695–717.